Induction: Gauss 5 y/o

$$1 + 2 + \cdots + 100 = S$$

$$100 + 99 + \cdots + 1 = S$$

$$\underbrace{101 + 101 \cdots + 101}_{10} = 2S = 101 \cdot 100$$

$$S = \frac{101 \cdot 100}{2}$$

WTP: $\forall n \in \mathbb{Z}_{>0}$, $\boxed{1 + 2 + \dots + n = \sum_{i=1}^{n} i} = \frac{n(n+1)}{2}$

$P(n)$

$n = 1 \quad 1 = \frac{1(1+1)}{2} = 1$

$n = 2 \quad 1 + 2 = \frac{1 \cdot 2}{2} + 2 = 2\left(\frac{1}{2} + 1\right) = 2\left(\frac{3}{2}\right)$

$n = 3 \quad 1 + 2 + 3 = \frac{2(3)}{2} + 3 = 3\left(\frac{2}{2} + 1\right) = 3\left(\frac{4}{2}\right)$

$1 + 2 + 3 + 4 = \frac{3 \cdot 4}{2} + 4 = 4\left(\frac{3}{2} + 1\right) = 4\left(\frac{5}{2}\right)$

Proof: Proceed by induction. The statement is already true for $n=1$ b/c $\boxed{\boxed{1 = \frac{1(2)}{2}}}$ ✓.

"Assume $P(n)$"

Assume that we already know the statement for n. I.E assume that

this is $P(n)$.

$$1+2+\dots+n = n(n+1)/2.$$

Adding $n+1$ to both sides gives

$$1+2+\dots+n+(n+1) = \frac{n(n+1)}{2} + (n+1).$$

The LHS of this is the LHS of what we want to prove. The RHS is

$$(n+1)\left(\frac{n}{2}+1\right) = (n+1)\left(\frac{n+2}{2}\right) = \frac{(n+1)(n+1+1)}{2}. \quad (\ast\ast)$$

We conclude that $\forall n \in \mathbb{Z}_{>0}$,

$$1+2+\dots+n = \frac{n(n+1)}{2}. \ \blacksquare$$

$\ast$ to $\ast\ast$ is a proof that $P(n) \Rightarrow P(n+1)$, Let $P(n)$ be a statement which depends on some integer (usually pos) n.

(Ex: $P(n) =$ "$1+2+\cdots+n = \frac{n(n+1)}{2}$")

Goal: Prove $P(n)$ $\forall n \in \mathbb{Z}_{>0}$.

Step 1: Prove $P(1)$ "Base Case"

Step 2: Prove "$P(n) \implies P(n+1)$" "Inductive step"

"Induction" = $P(1) \wedge$ "$P(n) \implies P(n+1)$" $\implies$

$\forall n \in \mathbb{Z}_{>0}, P(n)$

$P(1), P(2), P(3), P(4), \ldots, P(a), P(a+1), \ldots$

Want to go: $P(n) \neq \boxed{\frac{n(n+1)}{2}}$ ← not "T or F"

$P(n) = \boxed{1+2+\cdots+n = \frac{n(n+1)}{2}}$

Note: n is just a variable.

$P(1) \wedge P(n) \Rightarrow P(n+1)$ "prove the following case"

$P(a) \Rightarrow P(a+1)$

$P(a-1) \Rightarrow P(a)$

<u>Variant 3</u>: $P(0) \wedge P(n) \Rightarrow P(n+1)$

$P(2) \wedge P(n) \Rightarrow P(n+1)$

$$\boxed{P(2) \wedge P(n) \Rightarrow P(n+2)}$$

$\Rightarrow \forall n \in \mathbb{E}_{\geq 0},\ P(n)$

$P(-1) \wedge P(n) \Rightarrow P(n-1)$

$\forall n \in \mathbb{Z}_{<0},\ P(n)$

$P(-3),\ P(-2),\ P(-1)$

Define: a sequence $a_1, a_2, \ldots, a_n, \ldots$

$$\boxed{\begin{aligned} a_1 &= 2 = 2^1 \\ a_n &= 2 \cdot a_{n-1} \end{aligned}}$$

"recursive defn"

$a_n = 2 \cdot a_{n-1}$ $\boxed{a_{n+1} = 2 \cdot a_n}$

$$a_2 = 2 \cdot a_1 = 2 \cdot 2 = 4 = 2^2$$
$$a_3 = 2 \cdot a_2 = 2 \cdot 4 = 8 = 2^3$$
$$a_4 = 2 \cdot a_3 = 2 \cdot 8 = 16 = 2^4$$

<u>Claim</u>: $\forall n \in \mathbb{Z}_{\geq 0}$, $\boxed{a_n = 2^n.}$

$P(n) = $ " $a_n = 2^n$ "

Proof: Proceed by induction.

Base case: $P(1)$ is "$a_1 = 2^1$", i.e., "$2 = 2$". This is true.

$(P(n) \Rightarrow P(n+1))$

Inductive step: Assume $P(n)$, i.e., $a_n = 2^n$. (WTP $P(n+1)$, i.e., $a_{n+1} = 2^{n+1}$) Then $a_{n+1} = 2 \cdot a_n$ by the defn. of a_n. Then $a_{n+1} = 2 \cdot 2^n = 2^{n+1}$.

Thus $P(n+1)$ is true. $\boxtimes$

Define: $a_1 = 0$

$$a_n = \sqrt{3 + 2a_{n-1}}$$

Prove: $\forall n \in \mathbb{Z}_{>0}$,

$a_n < 3$

$P(n) =$ "$a_n < 3$"

$a_1 = 0$

$a_2 = \sqrt{3 + 2 \cdot 0} = \sqrt{3}$

$a_3 = \sqrt{3 + 2 \cdot a_2} = \sqrt{3 + 2\sqrt{3}}$

$a_4 = \sqrt{3 + 2a_3} = \sqrt{3 + 2\sqrt{3 + 2\sqrt{3}}}$

Proof: Proceed by induction.

Base case: $P(1)$ is "$a_1 < 3$", i.e., "$0 < 3$". This is true.

I.S. Assume $P(n)$, i.e., "$a_n < 3$".

[WTP: $P(n+1)$, i.e., "$a_{n+1} < 3$"] Then, by defn,

$a_{n+1} = \sqrt{3 + 2a_n}$. Since $a_n < 3$,

$$a_{n+1} = \sqrt{3 + 2a_n} < \sqrt{3 + 2 \cdot 3} = \sqrt{3+6}$$

$$= \sqrt{9} = 3$$

Thus $a_{n+1} < 3$. $\boxtimes$

Claim: "$1+2+4+8+\cdots+2^{n-1}=2^n-1$" $\forall n\in\mathbb{Z}_{>0}$

Proof: Proceed by induction.

BC: $P(1)$ is "$1=2-1$". This is true.

IS: Assume $P(i)$, i.e., $1+2+\cdots+2^{i-1}=2^i-1$.

(WTP $P(i+1)$, i.e., $1+2+\cdots+2^{i+1-1}=2^{i+1}-1$)

$1+2+\cdots+2^{i-1}+2^i=2^{i+1}-1$

Adding 2^i to each side gives

$$1+2+\dots+2^{i-1}+2^i = 2^i-1+2^i.$$

The LHS is the LHS of $P(i+1)$.

The RHS is $2\cdot 2^i - 1 = 2^{i+1}-1$.

This is the RHS of $P(i+1)$. $\blacksquare$

Claim: $\forall n \in \mathbb{Z}_{\geq 0}$, $\underbrace{3 \mid 4^n - 1}_{P(n)}$.

Proof: Proceed by induction.

BC: $P(0)$ is "$3 \mid 4^0 - 1$", i.e., "$3 \mid 0$"

which is true.

__IS:__ Assume $P(n)$, i.e., $3 \mid 4^n - 1$.

(WTP $3 \mid 4^{n+1} - 1$) Then $\exists m \in \mathbb{Z}$ s.t.

$4^n - 1 = 3m$. Then $4^n = 3m+1$.

Then $4^{n+1} - 1 = 4^n \cdot 4 - 1 = (3m+1) \cdot 4 - 1$

$= 12m + 4 - 1 = 12m + 3 = 3(4m+1)$.

Thus $3 \mid 4^{n+1} - 1$ $\blacksquare$